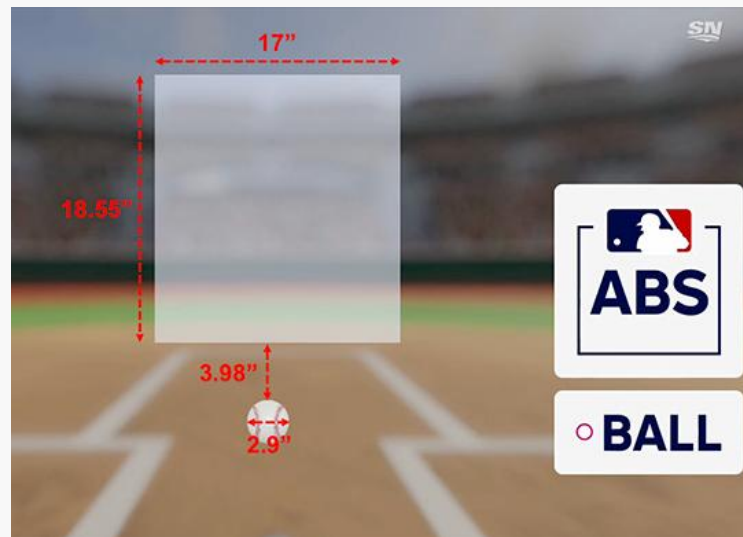

When Should You Challenge? Estimating ABS Challenge Value Using Matchup-Level RE288 Projections

Jordan Gottlieb, Ryan Ilan

Basic Idea

- Estimate run value of an ABS challenge for all 288 game states
 - Outs
 - Baserunners
 - Count
- Incorporated matchup data to refine run expectancy predictions
- Simplified results for use in game by players and coaches



Source: Sportsnet

Why

- What is the optimal way to use an ABS Challenge?
- Is it better to save your challenge for the best hitters?
- How does situation change the value of a challenge?
- Player vs Situation
- Best time in the count to challenge?

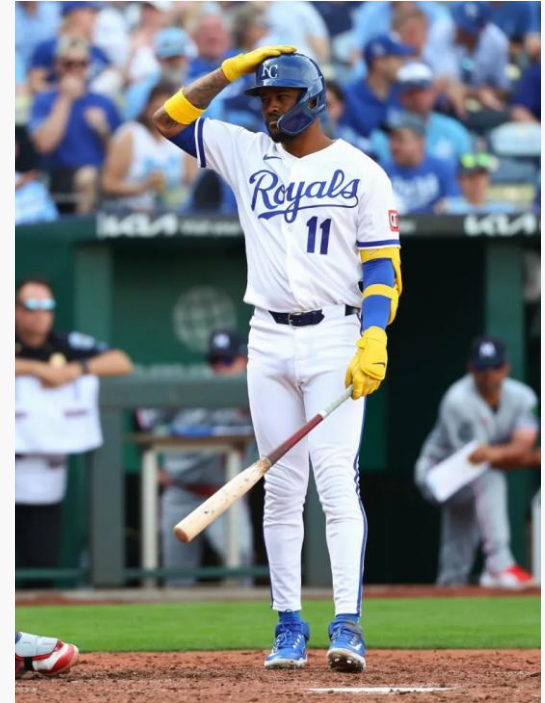


Image Credit: Mikayla Schlosser/MLB Photos
via Getty Images

Data

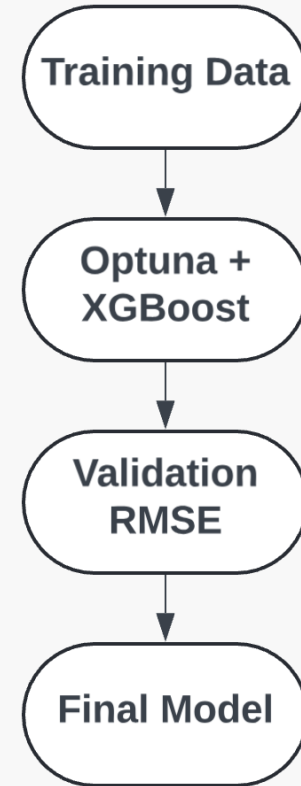
- Used data from 2023-2025 seasons
 - 2023-2024 Training set (~1.432 million pitches)
 - 2025 Validation set (~712,000 pitches)
- Sources:
 - Statcast
 - Baseball Reference
- Features:
 - Game State (Outs, Base Runner Position, Count)
 - Player Information (Handedness, PAs, ERA+, rBat+)



Source: Reddit (r/scorebug)

Model

- Algorithm: XGBoost
- Objective Function: Tweedie Loss
- Hyperparameter Tuning: Optuna
- Model Selection: Lowest validation RMSE



Results: Predicted RE288

Projected Run Expectancy Hitter A (L) vs Pitcher A (R)

rBAT+: 100 | ERA+: 100 | Avg Run Value: 0.867

outs bases	0-0	0-1	0-2	1-0	1-1	1-2	2-0	2-1	2-2	3-0	3-1	3-2
0 — — —	0.488	0.458	0.410	0.506	0.473	0.419	0.540	0.516	0.458	0.610	0.587	0.537
0 1B — —	0.840	0.797	0.740	0.861	0.817	0.758	0.903	0.868	0.808	1.003	0.959	0.905
0 — 2B —	0.962	0.915	0.813	0.987	0.933	0.825	1.031	0.989	0.870	1.140	1.095	0.989
0 1B 2B —	1.411	1.345	1.223	1.444	1.368	1.247	1.501	1.434	1.306	1.659	1.572	1.452
0 — — 3B	1.054	0.992	0.911	1.084	1.014	0.922	1.125	1.064	0.971	1.222	1.161	1.071
0 1B — 3B	1.570	1.493	1.401	1.608	1.518	1.416	1.650	1.563	1.471	1.843	1.740	1.627
0 — 2B 3B	1.920	1.816	1.647	1.967	1.845	1.662	2.015	1.899	1.712	2.170	2.049	1.863
0 1B 2B 3B	2.524	2.394	2.196	2.590	2.430	2.219	2.657	2.489	2.287	2.983	2.781	2.546
1 — — —	0.288	0.269	0.244	0.297	0.276	0.248	0.314	0.297	0.268	0.356	0.340	0.315
1 1B — —	0.507	0.478	0.450	0.520	0.488	0.459	0.540	0.512	0.485	0.595	0.563	0.538
1 — 2B —	0.598	0.560	0.501	0.613	0.569	0.507	0.637	0.601	0.533	0.708	0.670	0.609
1 1B 2B —	0.890	0.832	0.763	0.911	0.846	0.776	0.943	0.884	0.812	1.032	0.964	0.895
1 — — 3B	0.670	0.630	0.577	0.688	0.641	0.582	0.709	0.664	0.606	0.773	0.729	0.670
1 1B — 3B	1.010	0.956	0.900	1.035	0.968	0.906	1.056	0.988	0.933	1.170	1.096	1.025
1 — 2B 3B	1.281	1.199	1.082	1.314	1.216	1.090	1.341	1.248	1.117	1.448	1.353	1.219
1 1B 2B 3B	1.688	1.579	1.447	1.734	1.599	1.460	1.775	1.638	1.501	1.972	1.819	1.658
2 — — —	0.118	0.103	0.081	0.125	0.110	0.087	0.140	0.128	0.103	0.158	0.146	0.123
2 1B — —	0.243	0.218	0.185	0.251	0.227	0.193	0.273	0.252	0.218	0.297	0.275	0.243
2 — 2B —	0.285	0.259	0.210	0.296	0.270	0.218	0.320	0.298	0.244	0.351	0.328	0.280
2 1B 2B —	0.467	0.431	0.368	0.479	0.443	0.378	0.508	0.476	0.414	0.547	0.509	0.452
2 — — 3B	0.295	0.266	0.223	0.309	0.280	0.233	0.332	0.306	0.258	0.360	0.335	0.290
2 1B — 3B	0.485	0.445	0.394	0.499	0.458	0.403	0.523	0.484	0.433	0.574	0.533	0.476
2 — 2B 3B	0.598	0.553	0.468	0.619	0.572	0.479	0.646	0.601	0.511	0.692	0.647	0.561
2 1B 2B 3B	0.819	0.763	0.671	0.841	0.781	0.680	0.869	0.808	0.718	0.956	0.886	0.789

Projected Run Expectancy Hitter B (L) vs Pitcher A (R)

rBAT+: 190 | ERA+: 100 | Avg Run Value: 0.983

outs bases	0-0	0-1	0-2	1-0	1-1	1-2	2-0	2-1	2-2	3-0	3-1	3-2
0 — — —	0.564	0.530	0.478	0.580	0.545	0.496	0.611	0.586	0.521	0.684	0.661	0.607
0 1B — —	0.943	0.897	0.828	0.962	0.917	0.845	1.000	0.964	0.893	1.107	1.061	0.997
0 — 2B —	1.120	1.066	0.947	1.146	1.085	0.955	1.184	1.136	0.998	1.298	1.249	1.125
0 1B 2B —	1.641	1.568	1.410	1.673	1.593	1.433	1.725	1.650	1.491	1.895	1.800	1.647
0 — — 3B	1.191	1.116	1.020	1.223	1.140	1.030	1.260	1.187	1.076	1.371	1.299	1.190
0 1B — 3B	1.774	1.679	1.557	1.817	1.709	1.574	1.854	1.750	1.624	2.076	1.953	1.805
0 — 2B 3B	2.223	2.098	1.873	2.281	2.133	1.888	2.326	2.184	1.937	2.506	2.361	2.107
0 1B 2B 3B	2.983	2.824	2.537	3.066	2.871	2.562	3.126	2.918	2.628	3.507	3.263	2.925
1 — — —	0.324	0.302	0.277	0.333	0.310	0.281	0.347	0.328	0.299	0.388	0.371	0.346
1 1B — —	0.553	0.520	0.489	0.563	0.529	0.496	0.582	0.550	0.521	0.635	0.601	0.573
1 — 2B —	0.675	0.631	0.563	0.690	0.641	0.568	0.711	0.668	0.593	0.777	0.736	0.666
1 1B 2B —	0.997	0.932	0.844	1.017	0.945	0.855	1.047	0.979	0.891	1.132	1.057	0.971
1 — — 3B	0.744	0.694	0.634	0.763	0.706	0.638	0.782	0.727	0.660	0.848	0.796	0.727
1 1B — 3B	1.116	1.046	0.975	1.143	1.060	0.982	1.160	1.077	1.005	1.283	1.193	1.103
1 — 2B 3B	1.450	1.347	1.196	1.487	1.367	1.203	1.513	1.398	1.231	1.624	1.511	1.334
1 1B 2B 3B	1.937	1.799	1.613	1.992	1.825	1.626	2.028	1.858	1.666	2.240	2.054	1.829
2 — — —	0.141	0.122	0.100	0.148	0.130	0.106	0.164	0.148	0.122	0.182	0.167	0.144
2 1B — —	0.278	0.248	0.214	0.286	0.257	0.222	0.306	0.281	0.248	0.332	0.305	0.274
2 — 2B —	0.334	0.302	0.250	0.345	0.313	0.257	0.367	0.341	0.285	0.398	0.371	0.322
2 1B 2B —	0.539	0.496	0.426	0.551	0.509	0.436	0.576	0.538	0.472	0.616	0.573	0.511
2 — — 3B	0.339	0.303	0.257	0.353	0.317	0.267	0.377	0.344	0.292	0.407	0.377	0.328
2 1B — 3B	0.551	0.500	0.443	0.567	0.515	0.454	0.589	0.540	0.482	0.646	0.596	0.532
2 — 2B 3B	0.688	0.632	0.534	0.711	0.652	0.545	0.738	0.683	0.578	0.788	0.734	0.632
2 1B 2B 3B	0.952	0.884	0.769	0.979	0.904	0.778	1.004	0.927	0.816	1.100	1.016	0.893

Results: Challenge Run Value

Correct Challenge Value Hitter C (L) vs Pitcher A (R)

rBAT+: 50 ERA+: 100 | Avg Challenge Value: 0.293

outs bases	0-0	0-1	0-2	1-0	1-1	1-2	2-0	2-1	2-2	3-0	3-1	3-2
0 — — —	0.044	0.073	0.112	0.064	0.106	0.157	0.085	0.125	0.232	0.229	0.279	0.511
0 1B — —	0.061	0.096	0.215	0.089	0.131	0.280	0.127	0.149	0.375	0.430	0.484	0.859
0 — 2B —	0.067	0.119	0.211	0.098	0.165	0.260	0.142	0.214	0.378	0.300	0.395	0.773
0 1B 2B —	0.098	0.150	0.331	0.136	0.191	0.397	0.221	0.255	0.547	0.880	0.985	1.532
0 — — 3B	0.089	0.120	0.228	0.106	0.153	0.285	0.153	0.190	0.386	0.371	0.459	0.845
0 1B — 3B	0.110	0.141	0.361	0.131	0.167	0.427	0.268	0.270	0.583	0.722	0.836	1.419
0 — 2B 3B	0.141	0.193	0.362	0.164	0.229	0.411	0.268	0.330	0.569	0.428	0.600	1.169
0 1B 2B 3B	0.182	0.227	0.499	0.216	0.255	0.562	0.482	0.480	0.828	0.747	0.961	1.789
1 — — —	0.025	0.037	0.125	0.035	0.052	0.148	0.052	0.067	0.192	0.150	0.173	0.365
1 1B — —	0.038	0.048	0.209	0.052	0.064	0.244	0.077	0.077	0.296	0.302	0.326	0.622
1 — 2B —	0.046	0.066	0.216	0.064	0.092	0.244	0.098	0.128	0.319	0.204	0.257	0.576
1 1B 2B —	0.073	0.084	0.296	0.096	0.110	0.337	0.144	0.147	0.425	0.667	0.725	1.150
1 — — 3B	0.055	0.070	0.282	0.064	0.086	0.311	0.105	0.121	0.376	0.253	0.309	0.685
1 1B — 3B	0.074	0.081	0.409	0.085	0.092	0.443	0.174	0.162	0.537	0.544	0.612	1.148
1 — 2B 3B	0.104	0.126	0.479	0.118	0.148	0.506	0.198	0.230	0.615	0.304	0.424	1.039
1 1B 2B 3B	0.142	0.147	0.615	0.163	0.167	0.652	0.327	0.312	0.818	0.866	1.011	1.829
2 — — —	0.017	0.025	0.061	0.026	0.036	0.076	0.024	0.034	0.091	0.079	0.098	0.189
2 1B — —	0.029	0.039	0.144	0.043	0.055	0.170	0.040	0.047	0.191	0.187	0.213	0.404
2 — 2B —	0.033	0.056	0.174	0.046	0.074	0.200	0.047	0.073	0.231	0.131	0.172	0.404
2 1B 2B —	0.048	0.072	0.318	0.067	0.094	0.355	0.072	0.088	0.393	0.281	0.331	0.725
2 — — 3B	0.038	0.053	0.181	0.048	0.068	0.207	0.049	0.068	0.236	0.130	0.170	0.405
2 1B — 3B	0.052	0.064	0.326	0.063	0.077	0.356	0.086	0.090	0.397	0.278	0.328	0.725
2 — 2B 3B	0.061	0.099	0.404	0.072	0.115	0.436	0.090	0.127	0.486	0.161	0.238	0.725
2 1B 2B 3B	0.078	0.104	0.594	0.088	0.116	0.629	0.151	0.159	0.701	0.937	1.023	1.725

Correct Challenge Value Hitter B (L) vs Pitcher A (R)

rBAT+: 190 ERA+: 100 | Avg Challenge Value: 0.357

outs bases	0-0	0-1	0-2	1-0	1-1	1-2	2-0	2-1	2-2	3-0	3-1	3-2
0 — — —	0.051	0.068	0.162	0.066	0.100	0.198	0.098	0.139	0.283	0.282	0.336	0.619
0 1B — —	0.065	0.088	0.292	0.083	0.119	0.340	0.143	0.169	0.444	0.580	0.644	1.088
0 — 2B —	0.080	0.138	0.282	0.099	0.179	0.324	0.162	0.251	0.450	0.392	0.516	0.966
0 1B 2B —	0.106	0.183	0.435	0.132	0.217	0.494	0.245	0.309	0.650	1.182	1.336	1.985
0 — — 3B	0.107	0.120	0.286	0.119	0.157	0.332	0.183	0.223	0.446	0.474	0.583	1.030
0 1B — 3B	0.139	0.151	0.458	0.145	0.176	0.508	0.327	0.330	0.689	1.030	1.178	1.867
0 — 2B 3B	0.183	0.260	0.438	0.193	0.296	0.487	0.322	0.424	0.657	0.622	0.876	1.533
0 1B 2B 3B	0.242	0.333	0.625	0.255	0.357	0.691	0.589	0.635	0.988	0.720	1.058	2.046
1 — — —	0.030	0.033	0.140	0.038	0.048	0.158	0.060	0.072	0.205	0.182	0.207	0.412
1 1B — —	0.044	0.041	0.218	0.052	0.054	0.243	0.084	0.080	0.295	0.396	0.424	0.719
1 — 2B —	0.059	0.077	0.235	0.070	0.100	0.260	0.109	0.143	0.332	0.261	0.331	0.664
1 1B 2B —	0.086	0.101	0.317	0.102	0.123	0.353	0.154	0.165	0.432	0.880	0.966	1.398
1 — — 3B	0.069	0.071	0.299	0.076	0.089	0.321	0.121	0.136	0.388	0.319	0.389	0.777
1 1B — 3B	0.097	0.085	0.431	0.100	0.096	0.455	0.206	0.188	0.553	0.744	0.834	1.386
1 — 2B 3B	0.139	0.171	0.516	0.146	0.195	0.543	0.226	0.280	0.646	0.426	0.603	1.249
1 1B 2B 3B	0.193	0.212	0.674	0.202	0.232	0.714	0.382	0.389	0.877	0.883	1.108	1.985
2 — — —	0.026	0.030	0.106	0.034	0.042	0.122	0.034	0.045	0.144	0.111	0.133	0.278
2 1B — —	0.038	0.043	0.222	0.049	0.059	0.248	0.050	0.058	0.274	0.233	0.265	0.539
2 — 2B —	0.043	0.064	0.257	0.054	0.083	0.285	0.057	0.087	0.322	0.167	0.217	0.539
2 1B 2B —	0.055	0.083	0.436	0.067	0.102	0.472	0.078	0.100	0.511	0.379	0.441	0.952
2 — — 3B	0.050	0.060	0.267	0.060	0.077	0.292	0.063	0.085	0.328	0.173	0.223	0.551
2 1B — 3B	0.067	0.072	0.454	0.074	0.087	0.482	0.106	0.114	0.532	0.357	0.420	0.952
2 — 2B 3B	0.079	0.118	0.545	0.086	0.138	0.578	0.105	0.156	0.632	0.218	0.320	0.952
2 1B 2B 3B	0.096	0.135	0.778	0.100	0.149	0.816	0.173	0.201	0.893	0.936	1.059	1.952

*Charts are intended to be used by either side of the ball, fielders or hitters

Results: Required Confidence

Confidence Thresholds Hitter C (L) vs Pitcher A (R)

rBAT+: 50 | ERA+: 100

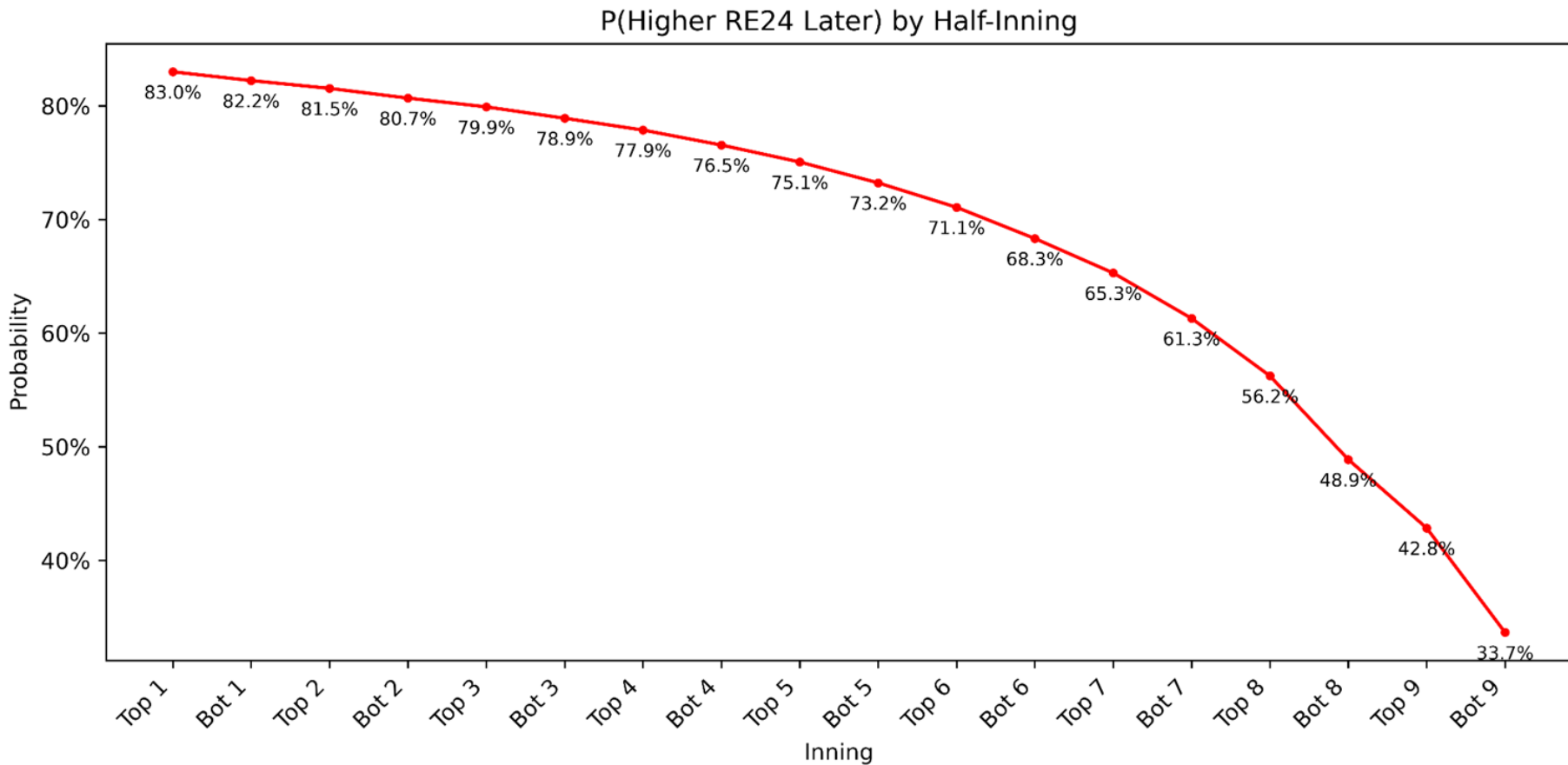
outs bases	0-0	0-1	0-2	1-0	1-1	1-2	2-0	2-1	2-2	3-0	3-1	3-2
0 — — —	81%	72%	62%	74%	64%	54%	69%	60%	44%	45%	40%	27%
0 1B — —	75%	66%	46%	68%	59%	40%	60%	56%	33%	30%	28%	18%
0 — 2B —	74%	61%	47%	66%	53%	42%	57%	47%	33%	38%	32%	19%
0 1B 2B —	66%	55%	36%	58%	49%	32%	46%	42%	25%	17%	16%	11%
0 — — 3B	68%	61%	45%	64%	55%	40%	55%	50%	33%	33%	29%	18%
0 1B — 3B	63%	57%	34%	59%	53%	30%	41%	41%	24%	20%	18%	12%
0 — 2B 3B	57%	49%	34%	53%	45%	31%	41%	36%	25%	30%	24%	14%
0 1B 2B 3B	51%	45%	27%	46%	42%	25%	28%	28%	18%	20%	16%	9%
1 — — —	88%	83%	60%	84%	78%	56%	78%	73%	49%	55%	52%	34%
1 1B — —	83%	79%	47%	78%	74%	43%	71%	71%	39%	38%	36%	23%
1 — 2B —	80%	74%	46%	74%	67%	43%	65%	59%	37%	48%	42%	24%
1 1B 2B —	72%	69%	39%	66%	63%	36%	56%	56%	30%	22%	20%	14%
1 — — 3B	77%	73%	40%	74%	69%	37%	64%	61%	33%	42%	38%	21%
1 1B — 3B	72%	70%	31%	69%	67%	30%	52%	53%	26%	25%	23%	14%
1 — 2B 3B	64%	60%	28%	61%	56%	27%	48%	45%	23%	38%	30%	15%
1 1B 2B 3B	57%	56%	23%	53%	53%	22%	36%	37%	19%	18%	16%	9%
2 — — —	91%	88%	75%	88%	84%	71%	89%	84%	67%	70%	66%	50%
2 1B — —	86%	83%	56%	81%	77%	52%	82%	80%	49%	50%	47%	32%
2 — 2B —	85%	77%	52%	80%	72%	48%	80%	72%	45%	59%	52%	32%
2 1B 2B —	80%	72%	37%	74%	66%	34%	72%	68%	32%	40%	36%	20%
2 — — 3B	83%	78%	51%	79%	73%	47%	79%	73%	44%	59%	52%	31%
2 1B — 3B	78%	75%	36%	75%	71%	34%	69%	67%	32%	40%	36%	20%
2 — 2B 3B	75%	65%	32%	72%	62%	30%	67%	59%	28%	54%	44%	20%
2 1B 2B 3B	71%	64%	24%	68%	62%	23%	55%	54%	21%	17%	15%	10%

Confidence Thresholds Hitter B (L) vs Pitcher A (R)

rBAT+: 190 | ERA+: 100

outs bases	0-0	0-1	0-2	1-0	1-1	1-2	2-0	2-1	2-2	3-0	3-1	3-2
0 — — —	79%	73%	54%	74%	65%	49%	65%	57%	40%	40%	36%	23%
0 1B — —	74%	68%	39%	69%	61%	35%	57%	52%	30%	24%	22%	15%
0 — 2B —	70%	57%	40%	65%	51%	37%	53%	43%	29%	32%	26%	16%
0 1B 2B —	64%	50%	30%	58%	46%	27%	43%	38%	22%	14%	12%	9%
0 — — 3B	64%	61%	39%	61%	54%	36%	50%	45%	29%	28%	24%	15%
0 1B — 3B	57%	55%	29%	56%	51%	27%	36%	36%	21%	15%	14%	9%
0 — 2B 3B	50%	42%	30%	49%	39%	28%	37%	31%	22%	23%	18%	11%
0 1B 2B 3B	43%	36%	23%	42%	34%	21%	24%	23%	16%	21%	15%	8%
1 — — —	86%	85%	57%	83%	80%	54%	76%	72%	48%	51%	47%	31%
1 1B — —	81%	82%	46%	78%	77%	43%	69%	70%	39%	32%	30%	21%
1 — 2B —	76%	71%	44%	73%	65%	42%	63%	57%	36%	42%	36%	22%
1 1B 2B —	69%	65%	37%	65%	60%	35%	55%	53%	30%	17%	16%	12%
1 — — 3B	73%	72%	38%	71%	68%	37%	61%	58%	32%	37%	32%	19%
1 1B — 3B	66%	69%	30%	65%	66%	29%	48%	50%	25%	20%	18%	12%
1 — 2B 3B	57%	52%	27%	56%	49%	26%	45%	40%	22%	30%	24%	13%
1 1B 2B 3B	49%	47%	22%	48%	45%	21%	33%	32%	18%	17%	14%	9%
2 — — —	88%	86%	64%	85%	81%	60%	85%	81%	56%	63%	58%	40%
2 1B — —	83%	81%	46%	79%	76%	43%	79%	76%	40%	44%	41%	26%
2 — 2B —	81%	74%	42%	78%	69%	40%	77%	68%	37%	53%	46%	26%
2 1B 2B —	77%	69%	30%	73%	65%	28%	71%	65%	27%	33%	30%	16%
2 — — 3B	79%	76%	41%	76%	71%	39%	75%	69%	36%	52%	46%	25%
2 1B — 3B	74%	72%	29%	72%	68%	28%	64%	62%	26%	34%	31%	16%
2 — 2B 3B	70%	61%	25%	68%	57%	24%	64%	54%	23%	46%	37%	16%
2 1B 2B 3B	66%	58%	19%	65%	56%	19%	52%	48%	17%	17%	15%	9%

Probability of Better Situations Occurring



Probability of Better Situations Occurring

Top 1st

Probability at least one greater RE24 situation occurs after the current PA
Average for Inning: 83.0%

	0 outs	1 out	2 outs
— — —	100.0%	100.0%	100.0%
1B — —	98.5%	100.0%	100.0%
— 2B —	89.9%	99.9%	100.0%
— — 3B	80.0%	98.1%	100.0%
1B 2B —	63.8%	94.9%	100.0%
1B — 3B	33.1%	84.8%	100.0%
— 2B 3B	20.1%	81.7%	100.0%
1B 2B 3B	0.0%	47.2%	99.9%

Top 5th

Probability at least one greater RE24 situation occurs after the current PA
Average for Inning: 75.1%

	0 outs	1 out	2 outs
— — —	99.9%	100.0%	100.0%
1B — —	89.5%	98.6%	100.0%
— 2B —	70.5%	97.7%	100.0%
— — 3B	58.3%	87.3%	100.0%
1B 2B —	42.7%	78.9%	100.0%
1B — 3B	18.8%	64.0%	100.0%
— 2B 3B	11.5%	60.2%	98.4%
1B 2B 3B	0.0%	27.9%	97.4%

Bot 9th

Probability at least one greater RE24 situation occurs after the current PA
Average for Inning: 33.7%

	0 outs	1 out	2 outs
— — —	47.7%	33.3%	27.2%
1B — —	45.6%	31.2%	26.2%
— 2B —	32.0%	22.8%	21.1%
— — 3B	20.7%	22.2%	18.1%
1B 2B —	39.5%	25.9%	22.0%
1B — 3B	14.3%	22.5%	16.4%
— 2B 3B	7.9%	25.9%	21.0%
1B 2B 3B	0.0%	0.0%	0.0%

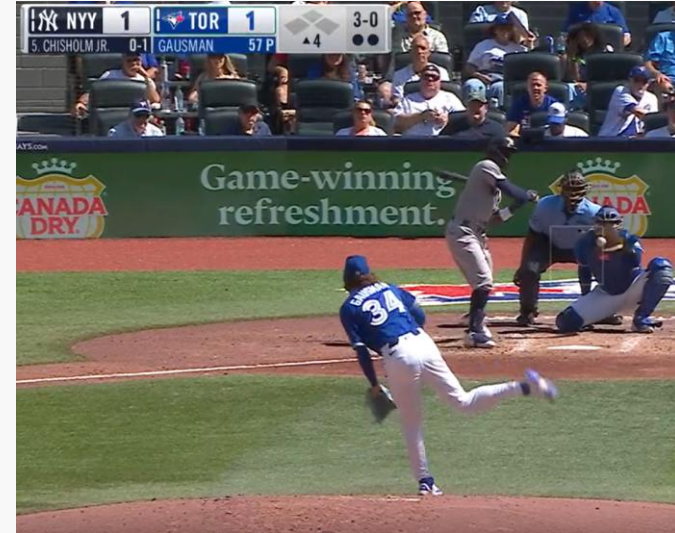
Game Time Implementation

Decision Matrix:	High Run expectancy game state + Late Inning	High Run expectancy game state + Early Inning	Low Run expectancy game state + Late Inning	Low Run expectancy game state + Early Inning
Late Count + Disproportionate Matchup	Lowest Confidence	Low Confidence	Less Confidence	More Confidence
Late Count + Neutral Matchup	Lowest Confidence	Low Confidence	Less Confidence	More Confidence
Early Count + Disproportionate Matchup	Less Confidence	Less Confidence	More Confidence	High Confidence
Early Count + Neutral Matchup	Less Confidence	Less Confidence	More Confidence	High Confidence

*Low confidence = greater challenge value

Game Time Implementation

- Post game audit sessions:
 - Players learn from experience
 - Situation specific numbers provided
 - Ask for their thought process
 - Strengthen team's understanding of challenge strategy



Source: YES Network



Reflection

- Future Work:
 - Using 2026 season data to include how many challenges are remaining to strategy
 - Use metrics to model probability calculation
- What we learned and how we would approach this differently:
 - Start simple before expanding on research
 - Complete one piece before working on a different idea

Thank You!

- Saberseminar
- California State University,
Northridge Anagnost College of
Engineering and Computer Science
- MLB
- Baseball Savant
- Baseball Reference



Link to Project Repository